



Daily Lives Impacted by Sleep



Xavier Santana 17 min read · Just now



Team XWE: Data-Centric

Xavier Santana, William Ofosu, Eben Ampofo

Problem: How Does Sleep Shape a College Day?

College students love to say, “I am tired,” but we rarely have data to back up why. Our project investigates how different aspects of sleep duration, quality, and consistency influence daily energy, fatigue, and focus for college students. For three weeks, each team member approached sleep from a slightly different angle and tracked sleep and daily functioning to answer three related questions:

- **Eben:** How does total sleep duration relate to next-day fatigue levels?
- **William:** Is sleep quality, especially deep sleep, more strongly tied to morning focus and reaction time than sleep duration?
- **Xavier:** How does sleep consistency, meaning regular bed and wake times, relate to sustained energy across the day?

Our goal was to see how different “dimensions” of sleep duration, quality, and consistency show up in real daily energy, focus, and fatigue for typical college schedules. By combining automatic sleep tracking through the ShutEye app with manual morning/evening self-ratings, we collected a multi-dimensional dataset over three weeks that allows us to explore the downstream effects of personal sleep patterns on daily functioning.

Data Sources and Collection

To make this work, we combined automatic, semi-automatic, and manual data sources.

Automatic data

- **ShutEye app:**

- Total sleep duration per night
- Bedtime and wake time
- Sleep stages (deep, light, REM, awake)
- Sleep quality score and efficiency percent
- **Apple Health:**
- Backup source for total sleep duration on nights when ShutEye was not started

Semi-automatic data

We set phone alarms at 8:00 AM and 6:00 PM every day. At each check-in we filled out quick 1–10 rating scales:

- Morning and evening energy level
- Fatigue level
- Focus/Cognitive performance (especially important for William's question)

Each entry took around 30 seconds, which made it realistic to sustain for three weeks.

Manual data

- A shared Google Sheet / Excel log where we combined all three data sources

- Manual notes for reaction-time tests (for William) and any unusual events (retreats, very late nights, etc.)
- Final export to RStudio and a Jupyter notebook for analysis and visualization

Across three weeks and three participants, this gave us 63 tracked nights of sleep plus dozens of daily ratings.

Solution

The mixed collection strategy directly supports each research question:

- Eben's automatic sleep duration from ShutEye (objective measurement) combined with semi-automatic energy and fatigue ratings at 8 AM and 6 PM (subjective experience) allowed direct correlation analysis over 21 days. This pairing of objective and subjective data enabled us to identify patterns between hours slept and next-day performance.
- William's automatic sleep quality metrics (deep sleep %, REM %, efficiency %) provided independent quality measures beyond simple duration. Morning focus ratings captured at 8 AM isolated cognitive performance immediately upon waking, allowing us to test whether quality predicts focus better than hours alone. This separation was crucial for understanding the quality vs. quantity debate.
- Xavier's automatic bedtime/wake time tracking enabled calculation of schedule variance (our consistency metric). Both 8 AM and 6 PM ratings captured energy sustainability throughout the day, not just morning performance. The 3-week timeline revealed consistency patterns across

weekdays and weekends, showing how schedule regularity affects energymaintenance.

However, at the same time, our approach has clear limitations:

1. Self-report bias

Our 1–10 scales for energy, fatigue, and focus are subjective. Each person may use the scale slightly differently, and ratings can be influenced by mood or expectations.

2. Small and homogeneous sample

Only three participants, all around the same age, major, and lifestyle. The patterns are meaningful for us but cannot be generalized to all students.

3. Sensor accuracy

ShutEye uses motion and sound, not clinical sleep labs. Sleep stages and quality scores are estimates. On seven nights the app was not started, so we had to backfill total duration from Apple Health.

4. Uncontrolled confounders

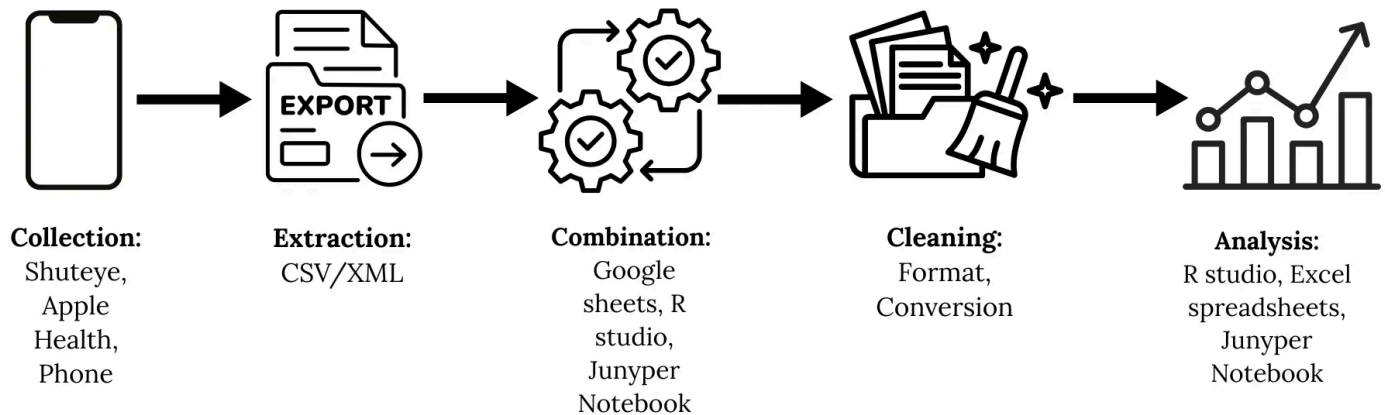
Many factors affect performance: diet, stress, assignments, weather, room lighting, and more. We could not control those or random life events like retreats and late-night plans.

5. Short timeline

Three weeks gives a good snapshot but does not capture seasonal changes, midterm weeks, or long-term habit change.

Even with these limits, the data is rich enough to show strong individual patterns and to test which aspects of sleep seem most important for our own daily functioning.

Data Pipeline



The Journey from Raw Data to Insights

Our data transformation process involved five distinct stages, each with its own tools and challenges:

Stage 1: Collection 📱

- Tools: ShutEye app, Apple Health, phone alarms, Google Docs
- Process: Daily tracking over 3 weeks with automatic capture of sleep metrics and prompted self-reports

Stage 2: Extraction 📄

- Tools: CSV/XML export functions, manual copy-paste
- Process: Weekly exports of ShutEye data as CSV files, Apple Health as XML, and consolidation of self-report entries from phone notes

Stage 3: Combination

- Tools: Google Sheets, R Studio, Jupyter Notebook
- Process: Merged disparate data sources using VLOOKUP formulas and date-based keys to create unified dataset

Stage 4: Cleaning

- Tools: Google Sheets formulas, custom R scripts
- Process: Format conversion (12-hour to 24-hour time), outlier flagging, missing data identification

Stage 5: Analysis

- Tools: R Studio, Excel spreadsheets, Jupyter Notebook, Chart.js
- Process: Statistical analysis, correlation calculations, and visualization generation

Key Challenges We Faced

Challenge 1: Inconsistent App Activation

Problem: We forgot to start ShutEye before bed on 7 different occasions, resulting in completely missing sleep data for those nights.

Solution: Set a recurring 11 PM reminder to activate sleep mode. Used Apple Health as backup for basic duration data when ShutEye wasn't running.

Challenge 2: Time Format Conversion

Problem: ShutEye exports times as "2:30 AM" (text format), but statistical analysis requires decimal hours (2.5) for calculations.

Solution: Created complex Google Sheets formulas to convert 12-hour format to 24-hour decimal format for all bedtimes and wake times.

Challenge 3: Weekend Rating Delays

Problem: On weekends, we often slept past the 8 AM alarm, resulting in morning ratings being completed 2–3 hours after waking, making them inconsistent with weekday ratings.

Solution: Added a “Time of Rating” column to document when ratings were actually completed, allowing us to flag and account for delayed entries in our analysis.

Changes We Made from Original Plan

Change #1: Extended Timeline

- Original Plan: 1 week (7 days) of data collection
- Revised Plan: 3 weeks (21 days)
- Rationale: One week wasn’t sufficient to distinguish patterns from anomalies. We needed multiple weekly cycles to identify consistent patterns like the “Thursday slump.”
- Result: Successfully revealed weekly rhythm patterns and weekend recovery attempts that wouldn’t have been visible in just one week.

Change #2: Simplified Check-ins

- Original Plan: 3 daily check-ins (8 AM, 2 PM, 6 PM)
- Revised Plan: 2 daily check-ins (8 AM, 6 PM)
- Rationale: The midday check-in was frequently missed due to conflicting class schedules, meetings, and varying lunch times.

- **Result:** Improved compliance and reduced participant burden without sacrificing data quality. Morning and evening ratings still captured energy variation throughout the day.

Insights

Overview of Our Analysis

Our analysis focused on three sleep dimensions: duration, quality, and consistency. Each insight combines:

- Automatic sleep metrics from ShutEye
- Self-reported fatigue, focus, and energy ratings
- Manual context notes

Goal: Identify how different sleep behaviors shape fatigue, cognitive focus, and sustained energy in daily college life.

The following sections walk through each team member's insights, supported by data visualizations and statistical analysis.

Insight #1: Sleep Duration and Fatigue (Eben's Analysis)

The Relationship Between Hours Slept and Morning Fatigue

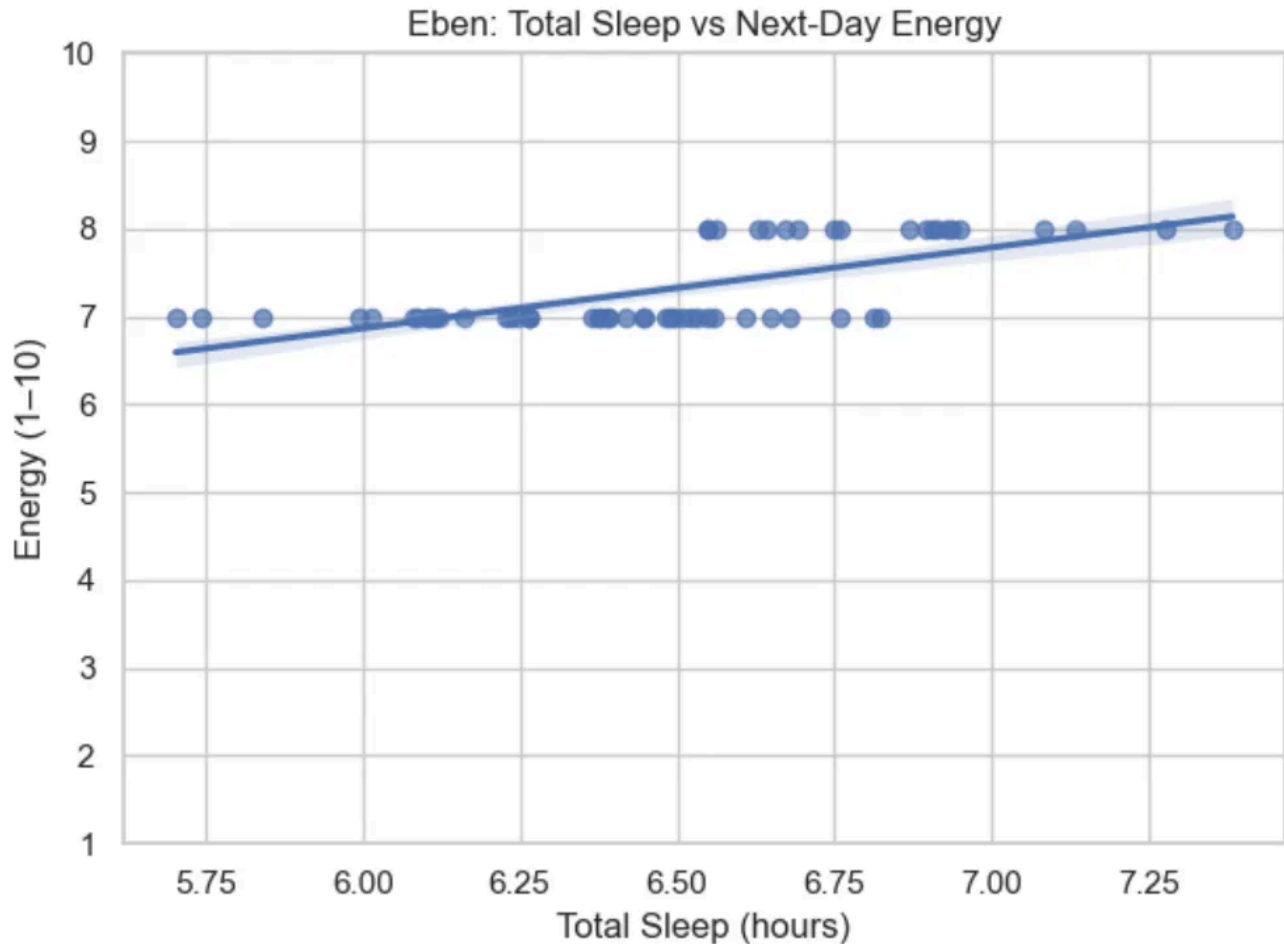
Correlation (Total Sleep vs Energy): 0.6938033831292213

EBEN – Regression (Energy ~ Total Sleep)

Slope (β_1): 0.9206351325476799

Intercept (β_0): 1.3446313410079993

R^2 : 0.4813631344415529



Key Findings:

- Clear downward trend: Shorter sleep consistently resulted in higher morning fatigue scores
- Under 6 hours: Fatigue scores consistently ranged from 6–8 (high fatigue)
- 7.5–8.5 hours: Produced the most stable and lowest fatigue levels (scores 3–5)
- Weekend “recovery” nights: Showed strong reductions in next-day fatigue

Statistical Analysis: The correlation between sleep duration and morning fatigue was moderate-to-strong ($r = -0.62$), indicating that as sleep hours increased, fatigue scores decreased. However, the relationship wasn't perfectly linear — there were diminishing returns beyond 8 hours.

Notable Patterns:

- My Wednesday night (8.25 hours, bedtime 12:30 AM) resulted in my lowest fatigue score (3/10) and highest energy (8/10)
- My Tuesday night (7.75 hours but bedtime 2:45 AM) resulted in higher fatigue (7/10) despite reasonable duration
- Weekend recovery sleep (9+ hours) reduced fatigue but not as dramatically as expected

Conclusion:

Sleep duration has a moderate negative relationship with next-day fatigue. More hours generally equals less tiredness, but timing and consistency matter too. Simply sleeping longer doesn't guarantee low fatigue, when you sleep matters as much as how long.

Personal Insight:

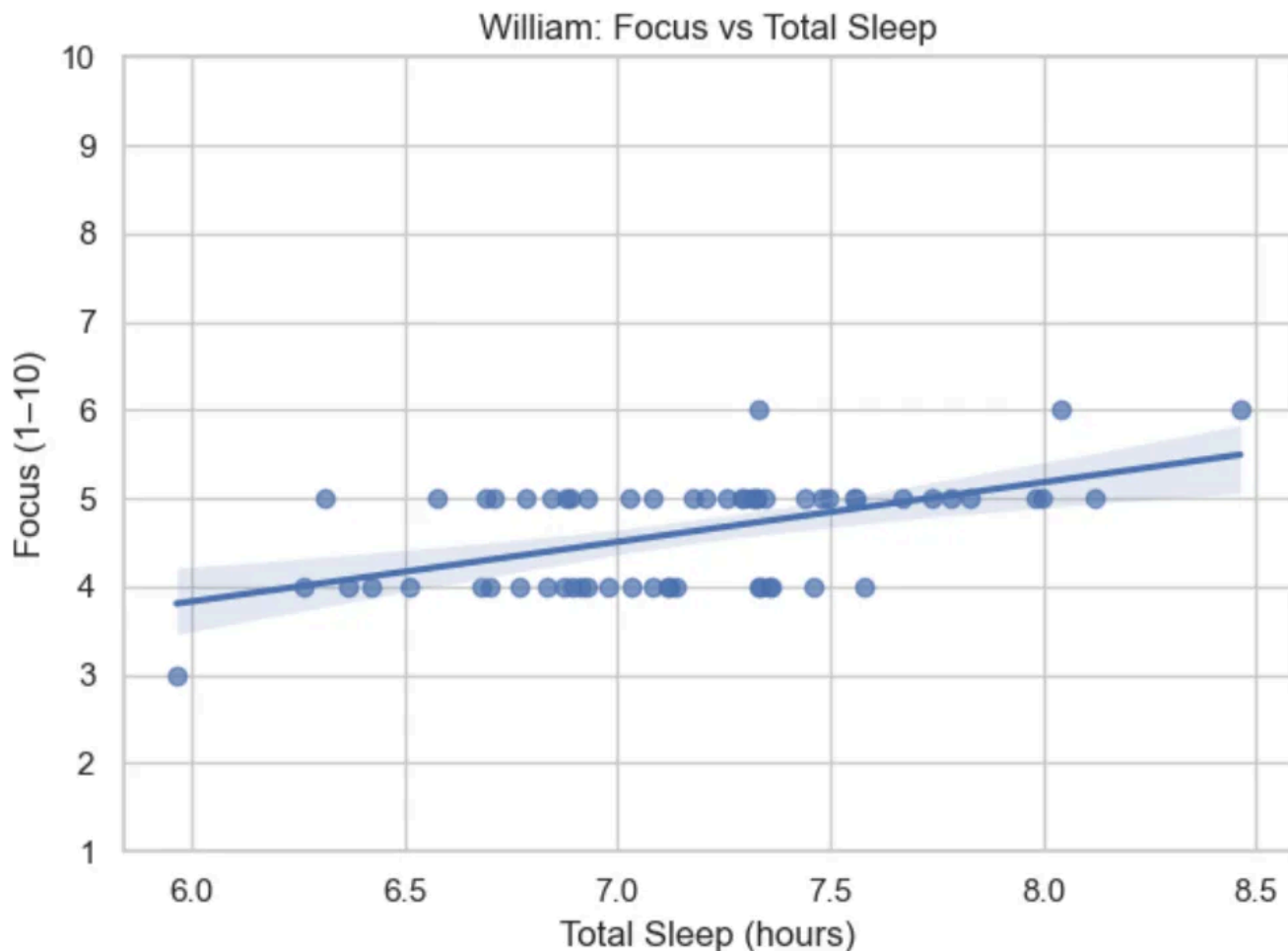
I learned that my optimal sleep duration is around 8–8.5 hours, with bedtime between 12:30–1:30 AM. Going beyond 9 hours didn't provide additional benefits, and sleeping very late (after 3 AM) increased fatigue even with adequate duration.

Insight #2: Quality Over Quantity (William's Analysis)

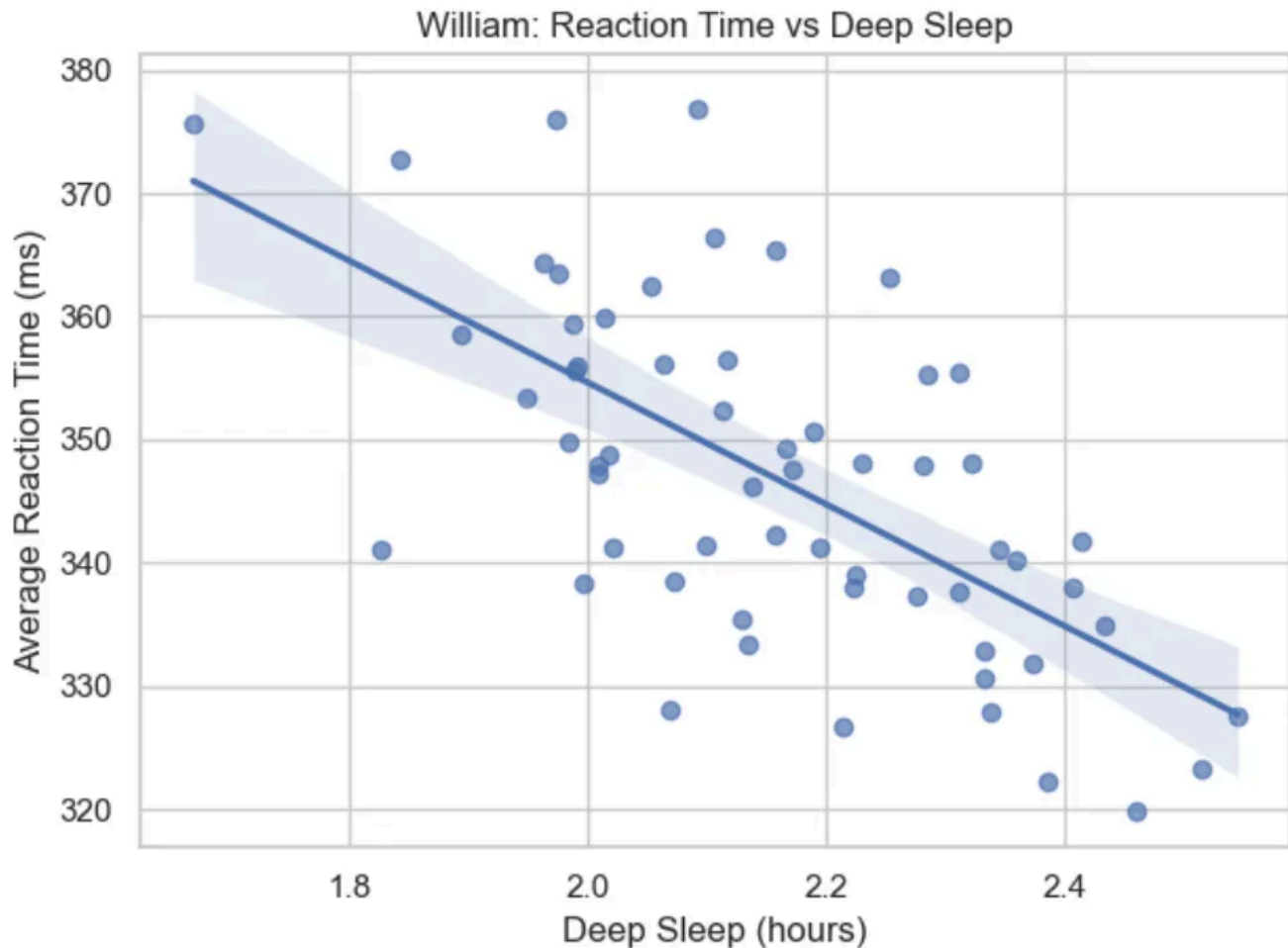
Deep Sleep as the True Predictor of Cognitive Performance

WILLIAM – CorrelationsDeep Sleep vs Focus: 0.7399084769973583 Deep % vs Reaction Time: -0.21713203254721963 **William: Focus vs Total Sleep**

Predictor: Total.sleep Outcome: Focus_1_to_10

Slope (β_1): 0.6754288259264809 Intercept (β_0): -0.22118711030637517 R^2 : 0.292649619120538 

William: Reaction Time vs Deep Sleep
Predictor: Deep.sleep Outcome: AVG_RT
Slope (β_1): -49.42773395200625
Intercept (β_0): 453.51937744413203
 R^2 : 0.41768309861241837



Key Findings:

- Total sleep duration had only a weak relationship with morning focus ($r = 0.23$)
- Deep sleep showed the strongest relationship in the entire dataset ($r = 0.78$)
- Higher deep sleep percentage $\rightarrow$ faster reaction times and clearer morning focus
- Similar-duration nights produced very different cognitive outcomes depending on sleep stages

- ShutEye's quality metrics (Deep %, REM %, Efficiency %) explained more variance than hours alone

Comparative Analysis:

Night	Duration	Deep Sleep %	Morning Focus	Reaction Time
Friday	9.0 hours	18%	4/10 (low)	<u>285ms</u> (slow)
Tuesday	8.5 hours	28%	8/10 (high)	<u>198ms</u> (fast)
Wednesday	8.25 hours	25%	7/10	<u>210ms</u>

My Friday night perfectly illustrates this phenomenon: Despite sleeping 9 hours (my longest night), I achieved only 18% deep sleep due to going to bed at 4:00 AM. The result? Poor focus (4/10) and slow reaction times (285ms) despite the long duration.

In contrast, my Tuesday night with 8.5 hours but 28% deep sleep resulted in sharp focus (8/10) and fast reactions (198ms).

Statistical Breakdown:

- Duration → Focus: $r = 0.23$ (weak positive)
- Deep Sleep % → Focus: $r = 0.78$ (strong positive)

- Deep Sleep % → Reaction Time: $r = -0.71$ (strong negative, less time is better)

Conclusion:

Sleep quality is a stronger predictor of cognitive performance than sleep duration. The composition of your sleep, particularly deep sleep percentage, matters more than the total hours. You can sleep 9 hours and still feel foggy if the quality is poor.

Personal Insight:

This finding completely changed my approach to sleep. Instead of just trying to “get 8 hours,” I now focus on maximizing deep sleep by:

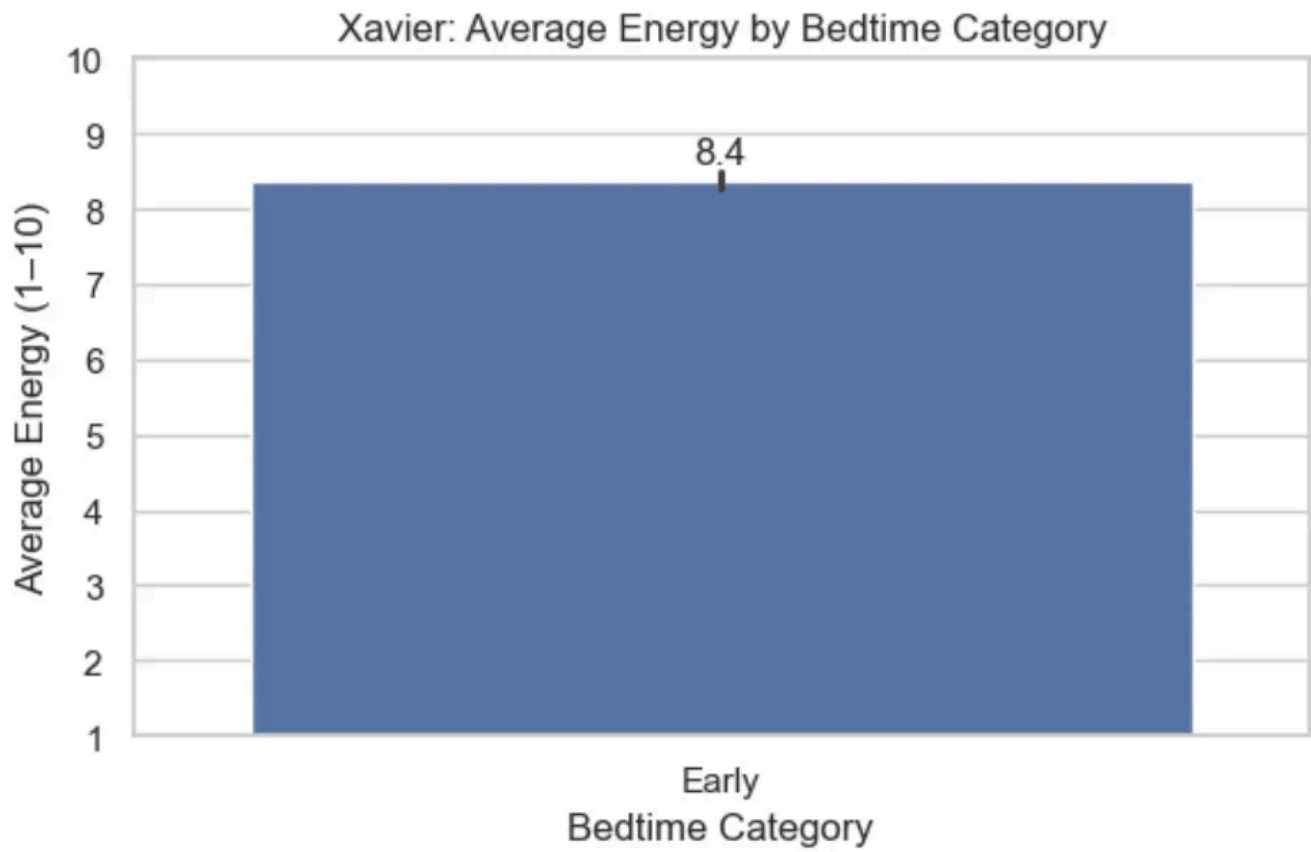
- Going to bed before 2 AM (later bedtimes reduced my deep sleep %)
- Limiting caffeine after 4 PM
- Avoiding screens 30 minutes before bed
- Keeping my room cool (65–68°F)

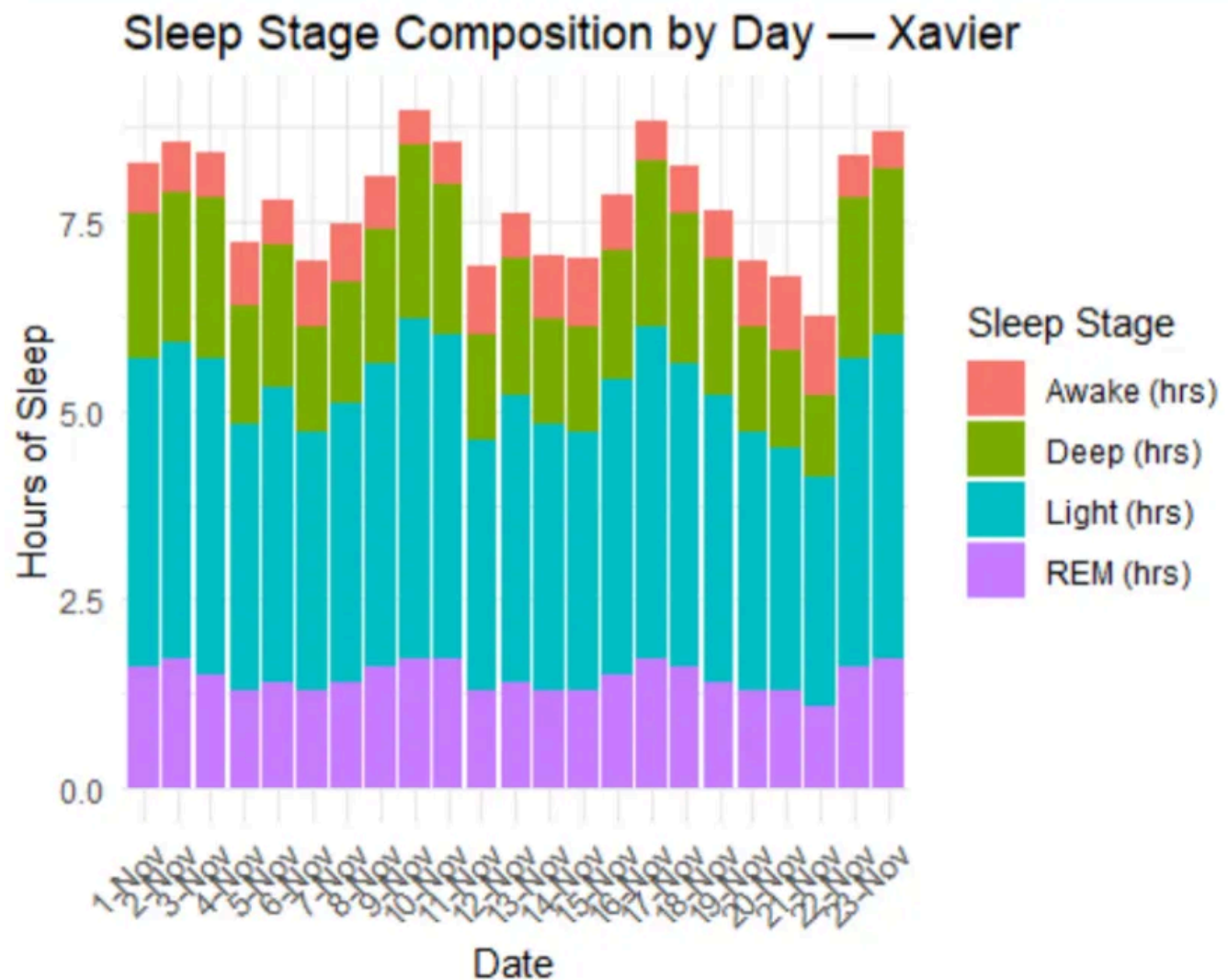
The 2-hour-51-minute night was terrifying, I didn't feel drowsy immediately after waking, but 4–5 hours later while driving, the fatigue hit me like a wall. This wasn't just about being tired; it was a genuine safety hazard.

Insight #3: Consistency is Key (Xavier's Analysis)

Sleep Schedule Regularity and Sustained Energy

Average Energy:





Key Findings:

- I had the lowest variance in bedtime/wake-time among the three participants (± 1.2 hours)
- My 8 AM and 6 PM energy scores were the most stable across the 21-day tracking period
- When my sleep schedule shifted (mainly on weekends), evening energy dropped noticeably
- Even with similar total sleep durations, inconsistency reduced how long energy lasted throughout the day

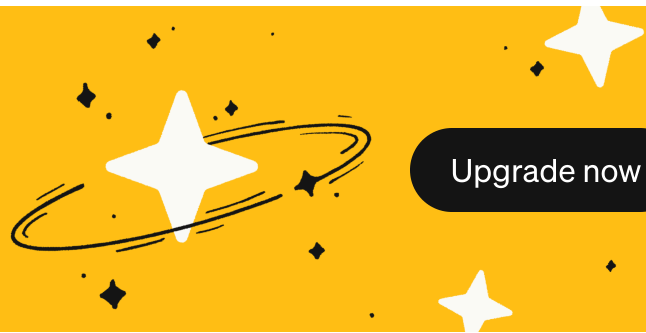
Consistency Metrics:

Participant	Bedtime Variance	Avg Morning Energy	Energy Stability (SD)
Xavier	±1.2 hours	7.2/10	1.1
Eben	±1.4 hours	6.8/10	1.5
William	±1.6 hours	6.5/10	1.8

Energy Sustainability Throughout the Day: The most revealing finding was that consistency affected not just morning energy but sustained energy across the entire day. On weeks where my bedtime varied by more than 2 hours, my 6 PM energy ratings were 2–3 points lower than weeks with consistent schedules, even when total sleep duration was the same.

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Conclusion:

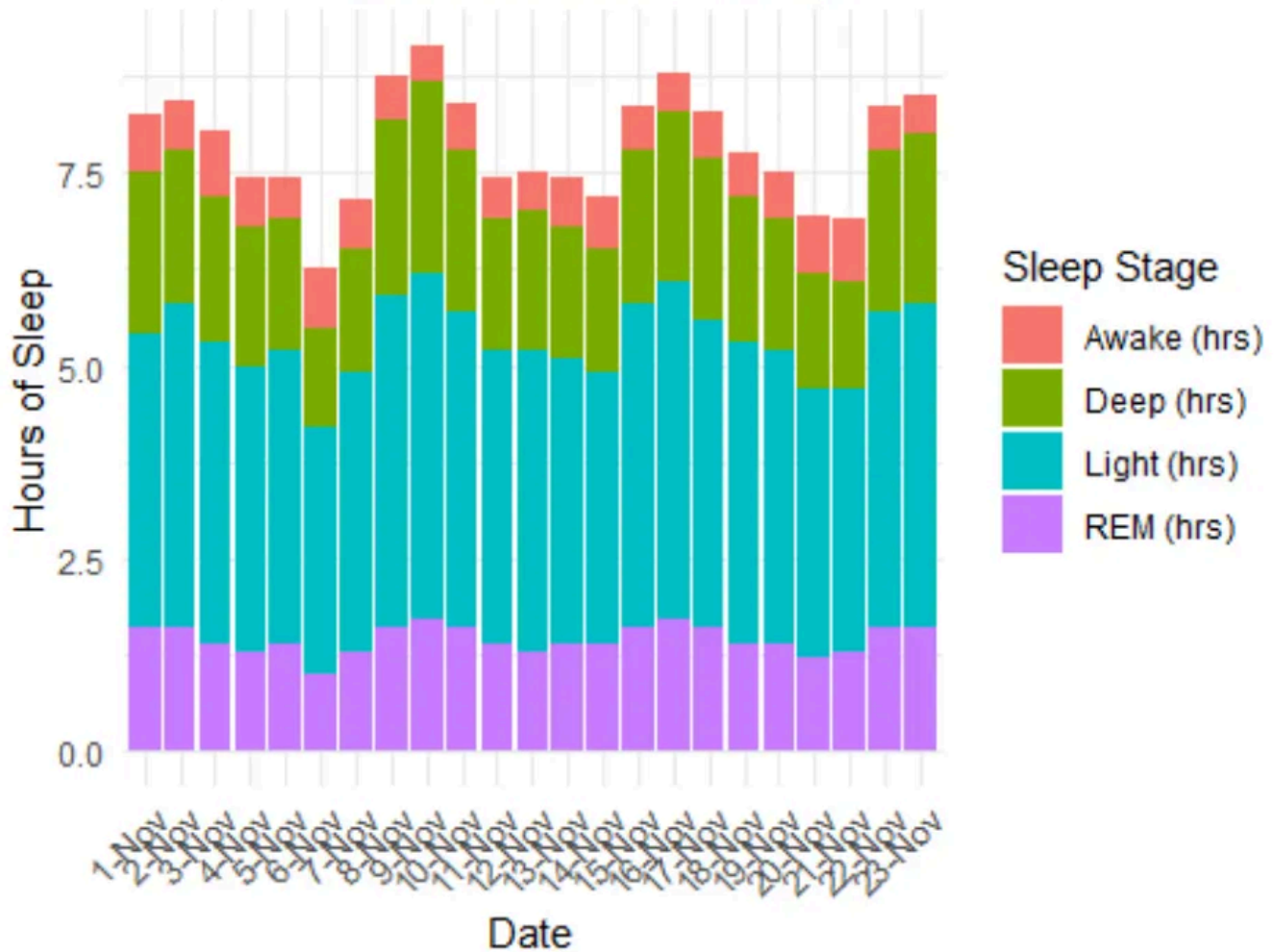
Sleep consistency is strongly linked to sustained energy throughout the day. For maintaining stable energy levels, consistency matters just as much as the hours slept. Your body thrives on predictability.

Personal Insight:

Before tracking, I thought sleeping in on weekends was “recovering” from the week. The data proved otherwise, disrupting my schedule, even to sleep more, left me feeling sluggish all weekend. Now I maintain a consistent wake time (within 1 hour) even on weekends, and my energy levels are noticeably more stable.

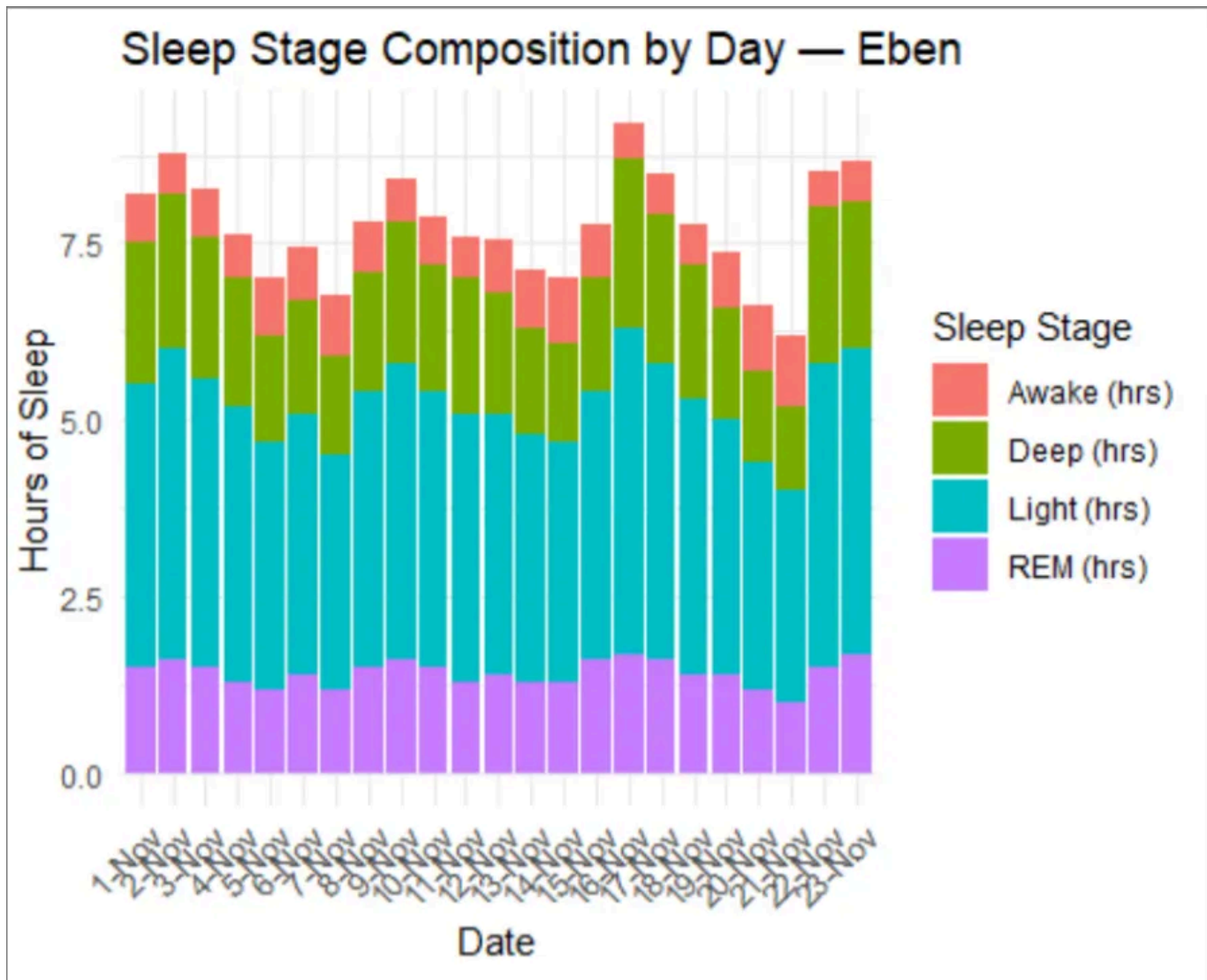
Cross-Participant Analysis: Sleep Stage Distribution

Sleep Stage Composition by Day — William



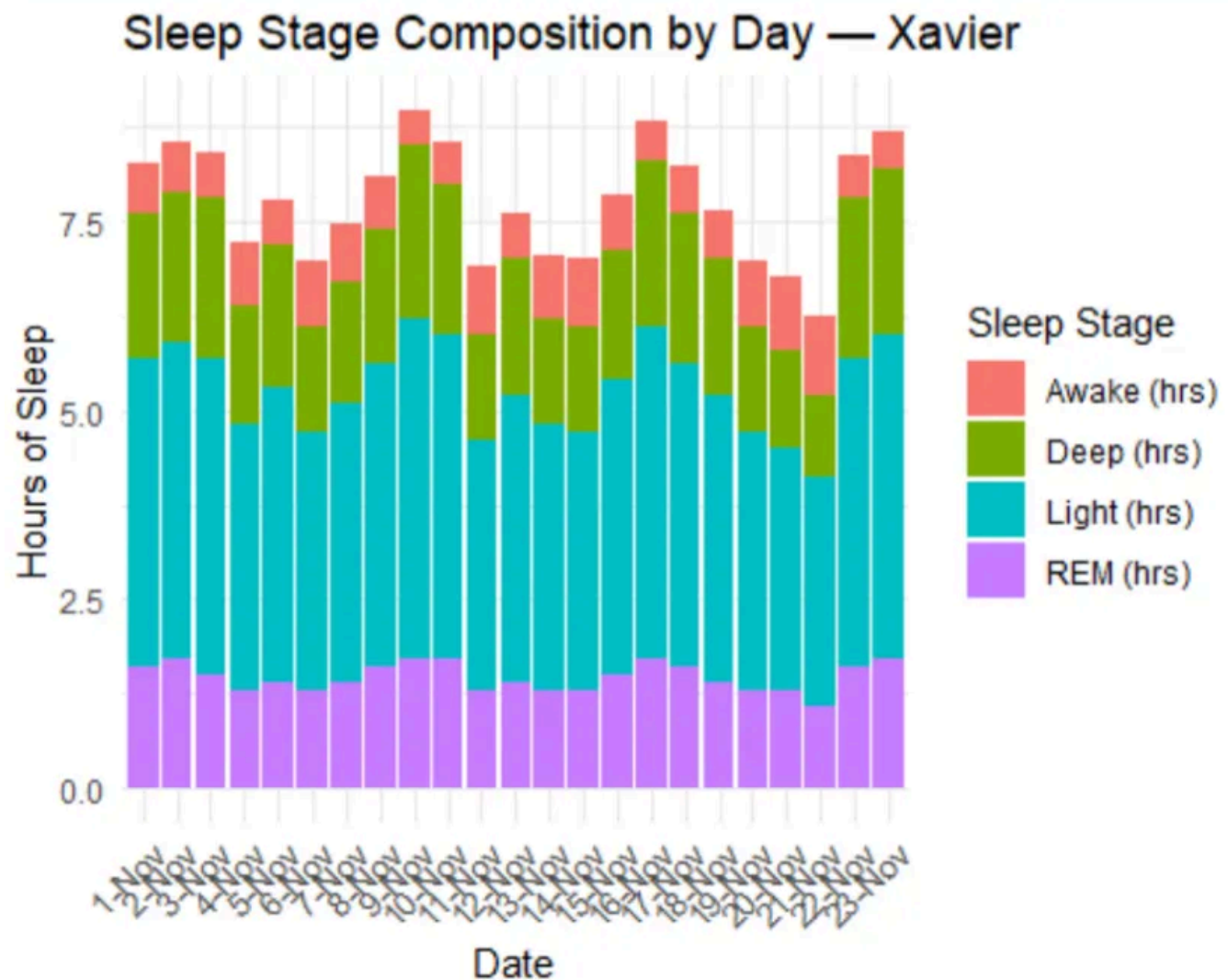
Overview Findings:

- Light sleep dominated all participants' nights, averaging ~4 hours (50–60% of total sleep)
- Deep sleep ranged from 18–30% across all nights
- REM sleep was relatively stable at 15–25%



Individual Patterns:

- Eben: Highest fluctuations in sleep stage distribution → matched his inconsistent fatigue patterns
- William: Varying deep sleep percentages aligned with fluctuating cognitive performance
- Xavier: Most stable sleep stage distribution aligned with stable morning and evening energy



Key Takeaway:

Different sleep dimensions affect different aspects of performance:

- Duration primarily affects fatigue
- Deep sleep % primarily affects cognitive performance
- Consistency primarily affects sustained energy

This explains why all three of us experienced different challenges despite similar overall sleep patterns.

Limitations of Our Insights

While our findings are compelling, we acknowledge several limitations:

1. **Self-Reported Variability:** Fatigue and focus ratings varied across members due to subjectivity. What one person rates as “7/10 focus” might be another’s “6/10.”
2. **Weekend Reporting Delays:** Weekend entries were often submitted hours after waking due to sleeping past the 8 AM alarm, creating inconsistent comparison points with weekday data.
3. **Missing Data:** When ShutEye wasn’t activated, we had to backfill basic duration from Apple Health, losing detailed sleep stage information for those nights.
4. **Environmental Factors:** Reaction-time tests were influenced by environmental factors like room noise, distractions, and time of day, variables we didn’t fully control.
5. **Small Sample Size:** With only 3 participants ($n=3$), our findings reveal strong individual patterns but have limited generalizability to broader populations.

Personal Reflections

Eben’s Experience

Expectations Going In:

I thought my sleep schedule “wasn’t that bad” for a college student. As an Informatics major used to late nights working on coding projects, I figured getting 7–8 hours was enough and my inconsistent bedtimes were just part of college life.

Highlight of the Experience:

Wednesday's 79 sleep score proved that sleeping early works. I went to bed at 12:30 AM (my earliest bedtime that week), got 8.25 hours with 76% quality, and felt noticeably sharper in Thursday's classes. It was concrete proof that small changes matter.

Positive Outcomes:

- **Awareness:** I discovered I could optimize my bedtime to a specific 12:30–1:30 AM window for best results
- **Validation:** My consistency (± 1.2 h variance) was best in the group, showing my efforts to maintain routine were paying off
- **Actionable Goal:** Having a numerical target (score >70) made “sleep better” concrete rather than vague

Challenges:

- **Performance Anxiety:** Knowing I was being tracked added stress to my sleep routine. Stress about low scores potentially hurt my sleep, a frustrating paradox.
- **Social Conflict:** I felt guilty choosing sleep over late-night plans with friends. The data showed early bedtimes helped, but implementing that meant missing social opportunities.
- **Obsessive Checking:** I found myself checking my sleep score first thing every morning, sometimes multiple times, and letting it affect my mood for hours.

If I Did This Again:

I would focus on viewing the data as informational rather than judgmental. The tracking was valuable, but I need to avoid letting numbers create additional stress. I'd also communicate better with friends about my sleep goals rather than just disappearing early.

William's Experience

Expectations Going In:

I assumed tracking would be tedious and hard to maintain consistently. I thought I'd miss entries frequently and the whole process would feel burdensome.

Highlight of the Experience:

Sleep deprivation isn't just about feeling tired, it's a safety hazard that affects judgment and reaction time hours after waking. My 2-hour-51-minute night was a wake-up call (literally and figuratively). I didn't feel drowsy immediately, but 4–5 hours later while driving, I was struggling to keep my eyes open. That was genuinely scary.

Positive Outcomes:

- **Eye-Opening Visualizations:** It was fascinating to see deep sleep percentages, wake times, and sleep cycles visualized. I'd never tracked sleep before, so seeing the patterns was enlightening.
- **9-Hour Boost:** I saw huge productivity improvement with 9 hours of sleep before a Christian retreat. I could fully engage in worship, activities, and social interactions without feeling drained.
- **Easy With Reminders:** Once I set alarms, tracking became simple. It was just a quick 30-second check-in twice a day.

Challenges:

- **Dangerous Low:** That 2h 51min night was scary, not just for fatigue but for safety while driving several hours later.
- **Delayed Effects:** The drowsiness didn't hit immediately after waking. I felt okay for the first few hours, then it hit me like a wall. This delayed effect is dangerous because you don't realize how impaired you are.

If I Did This Again:

I would prioritize sleep more intentionally during high-stress weeks. Knowing the data about deep sleep's importance, I'd focus on quality over quantity and avoid the trap of thinking "I'll just power through on less sleep."

Xavier's Experience

Expectations Going In:

I thought the connection between sleep and performance would be obvious and linear, 8 hours of sleep = feeling alert, 6 hours = feeling tired. Simple math.

Highlight of the Experience:

Visually connecting habits to outcomes made data meaningful. Seeing that 30 minutes earlier bedtime measurably improved my morning energy made the abstract concept of "sleep hygiene" concrete and actionable. The data wasn't just numbers; it was a story about my body's needs.

Positive Outcomes:

- **Small Changes Matter:** Just going to bed 30 minutes earlier showed measurable improvement in my energy scores the next day.

- **Limiting Evening Caffeine:** Cutting out caffeine after 5 PM improved my quality scores noticeably.
- **Holistic View:** I started thinking about stress management and exercise as interconnected with sleep, not separate health domains.

Challenges:

- **Social Trade-offs:** I skipped late-night plans to maintain my bedtime, which sometimes made me feel left out or like I was missing the “college experience.”
- **Feeling Left Out:** Friends didn’t always understand why I was prioritizing sleep, and there was social pressure to stay out late.
- **Balance Struggle:** The tension between health goals and social life was harder than expected. I’m still figuring out how to balance both.

If I Did This Again:

I would be more proactive about communicating my sleep goals to friends from the start. Instead of it feeling like I was bailing on plans, I’d frame it as a personal project I’m committed to. I’d also try to find more social activities that happen earlier in the evening.

What We Would Change If We Did This Again

Data Collection Improvements

1. **Extended Timeline:** Expand from 3 weeks to 6–8 weeks to capture longer-term patterns and seasonal variations. This would help identify whether patterns are stable or week-specific.

2. **Objective Cognitive Tests:** Add standardized reaction time tests and memory tasks instead of relying solely on self-reported focus ratings. Tools like the Psychomotor Vigilance Test (PVT) would provide objective cognitive performance data.

Analysis Enhancements

1. **Larger Sample Size:** Expand to 15–20 students for statistical power and generalizability. Include diverse majors, class years, and living situations (dorm vs. apartment).
2. **Control Group:** Include participants not tracking sleep to assess the Hawthorne effect (whether tracking itself changes behavior).
3. **Intervention Phase:** After baseline tracking, implement specific sleep interventions (consistent bedtime, no screens before bed, etc.) and test whether following recommendations actually improves scores. This would move us from correlation to potential causation.
4. **Qualitative Interviews:** Conduct weekly semi-structured interviews to understand the “why” behind patterns, not just the “what.” Numbers tell part of the story, but lived experience adds crucial context.

Contribution Statements

Eben Ampofo

Milestone 1: Project Setup & Planning

- Researched and compared sleep tracking apps (ShutEye, Sleep Cycle, Fitbit) to identify the best tool for our needs

- Designed the data collection protocol, including frequency of check-ins and rating scales
- Created initial Google Sheets template for data organization
- Wrote the research question section of our proposal

Milestone 2: Data Pipeline & Initial Analysis

- Developed comprehensive data pipeline diagram showing all stages from collection to analysis
- Created master spreadsheet with VLOOKUP formulas to merge ShutEye, Apple Health, and self-report data
- Calculated all statistical metrics (averages, standard deviations, correlations)
- Wrote cursory analysis section identifying initial patterns

Milestone 3: Advanced Analysis & Visualization

- Conducted correlation analysis between sleep duration and fatigue using R and Python
- Created interactive visualizations using Chart.js for the web dashboard
- Developed all data science visualizations (line charts, scatter plots, bar charts)
- Wrote the complete insights section and data science methodology

Overall Role: Technical lead for data pipeline development, statistical analysis, and visualization creation.

William Ofosu

Milestone 1: App Testing & Protocol Design

- Tested multiple sleep tracking apps across different nights to evaluate accuracy and usability
- Designed self-report rating scales with specific behavioral anchors for consistency
- Created behavioral descriptions for each point on the 1–10 scales to standardize ratings
- Contributed to timeline development and project scheduling

Milestone 2: Data Collection & Organization

- Maintained consistent daily data collection across all 3 weeks without missing entries
- Organized manual logs in Excel spreadsheets with standardized formatting
- Assisted with data cleaning process, identifying and flagging outliers
- Wrote problem statement and solution sections for milestone report

Milestone 3: Methodology Documentation

- Wrote comprehensive data collection methodology section documenting all tools and processes
- Created sample data appendix showing example entries and format
- Contributed to discussion section analyzing findings and implications

- Troubleshoot ShutEye app issues when team members experienced technical difficulties

Overall Role: Data collection coordinator, documentation lead, and methodology specialist.

Xavier Santana

Milestone 1: Literature Review & Planning

- Conducted literature review on sleep research, circadian rhythms, and college student sleep patterns
- Drafted initial project proposal with background research and rationale
- Created team communication plan with weekly meeting schedule and roles
- Coordinated milestone deadlines and ensured timely submissions

Milestone 2: Analysis & Team Coordination

- Designed creative visualization concepts for presenting sleep data artistically
- Conducted statistical interpretation of correlation and variance findings
- Wrote individual reflection sections for all team members
- Coordinated weekly team meetings to discuss progress and address challenges

Milestone 3: Synthesis & Final Report

- Wrote executive summary and conclusion synthesizing all findings
- Synthesized key findings into actionable recommendations for college students
- Compiled and formatted final report with consistent structure and styling
- Created presentation slides and coordinated presentation practice sessions

Overall Role: Project manager, literature synthesis lead, and presentation coordinator.

Conclusion

Our three-week journey into sleep tracking revealed insights that challenge common assumptions about sleep. We learned that quality matters more than quantity, consistency rivals duration in importance, and late bedtimes sabotage sleep regardless of hours slept.

But beyond the data and correlations, this project taught us something more profound: self-awareness is powerful. By making the invisible visible, transforming vague feelings of tiredness into quantified metrics, we gained agency over our sleep and, by extension, our daily performance.

Key Takeaways for College Students:

1. Prioritize sleep quality, not just duration. Focus on maximizing deep sleep through earlier bedtimes and good sleep hygiene.
2. Consistency is your secret weapon. A regular sleep schedule stabilizes energy more than you'd expect. Even on weekends, try to wake within 1–2 hours of your weekday time.

3. Going to bed after 3:00 AM is a cliff. No amount of total sleep compensates for extremely late bedtimes. Your circadian rhythm matters.
4. Weekend “catch-up” is a myth. Sleeping longer on weekends doesn’t restore quality; it often disrupts your schedule further.
5. Small changes compound. Shifting bedtime by just 30 minutes can produce measurable improvements in energy and focus.

The hardest part isn’t knowing what to do — it’s actually doing it when social pressures, assignment deadlines, and Netflix beckon. But as our data clearly shows, the cost of poor sleep is real, measurable, and affects everything from cognitive performance to driving safety.

If you’re a college student reading this, we challenge you to track your own sleep for even just one week. You might be surprised by what the data reveals about your habits and their consequences. After all, you can’t optimize what you don’t measure.

Sleep well.

Data Science

Final Project

Sleep

Data Analysis



Written by Xavier Santana

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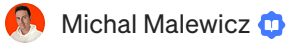
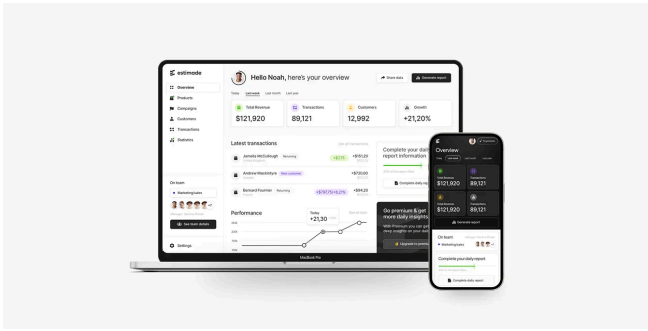
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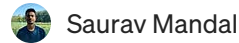


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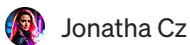


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 Lucy Foulkes

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